

POLARIS DC-1 — EXTERNAL WORKS, SECURITY AND FIBRE

Document Ref: PDC-SPEC-o8 Rev A
Project: Polaris DC-1 — Aviapolis Data Campus, Phase 1, Vantaa, Finland
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Date: 26 February 2026
Status: For Subcontract

Synthetic demonstration data. All parties, ratings and figures are fictional.

1. Scope

- 1.1 This specification covers the design, supply, installation, testing and commissioning of the external works at Polaris DC-1: site drainage and stormwater attenuation, perimeter security infrastructure, external lighting, landscaping, the site access road and haul road, and the two diverse fibre entry routes and meet-me room termination infrastructure.
- 1.2 The Subcontractor shall carry design responsibility for the means of compliance with PDC-ER-001 §1.2, §5 and §11, using PDC-BOD-001 §2 as the design intent for site layout to be carried forward unchanged unless agreed under the S1–S3 review process.
- 1.3 Excluded: the 110 kV overhead line and its exclusion zone (network operator, Employer's battery limit); carrier fibre tails beyond the meet-me room termination panels (Tenant/carrier-supplied, PDC-TA-001 §11); the district-heating offtaker's network beyond the boundary chamber (refer PDC-DH-001).
- 1.4 The external works shall be delivered in a west-to-east sequence matching the four Sections of PDC-ER-001 §7, with the single site haul road along gridline N and hoarding lines relocated progressively per PDC-BOD-001 §2.3, without interrupting the operation of any Section already taken over.

2. Standards

Reference	Application
National Building Code of Finland	Site drainage, external works, structural fire chapters for perimeter structures
Land Use and Building Act (132/1999)	Statutory design basis, planning permission conditions (PDC-PP-001)
Rescue Act (379/2011)	Fire service access routes, hydrant provision
SFS 6000	Low-voltage installations — external lighting, gate motors, ELV supplies to security equipment
EN 50600 series	Data centre facility design reference, telecom cabling pathway sections
EN 50173 series	Information technology — generic cabling systems, meet-me room and fibre pathway design reference
EN 50131 series	Intrusion and access control systems — perimeter detection
EN 62676 series	CCTV systems for use in security applications
RIL 126 / RIL 237 (Finnish drainage design guidance)	Stormwater attenuation design basis
Vantaa Water (municipal utility) technical	Foul and surface water connection standards

Where standards conflict, the more onerous requirement governs, per PDC-BOD-001 §1.2.

3. Equipment schedules

3.1 Perimeter security infrastructure

Item	Requirement
Perimeter fence	Full enclosure of the 4.8 ha plot, minimum 2.4 m, with anti-climb and anti-cut specification at the Engineer's approval
Intrusion detection	Continuous perimeter detection (fence-mounted or buried, subject to S2 selection), integrated with PDC-SPEC-07
Vehicle gate	ANPR-controlled, single Tikkurilantie access point
External lighting	Perimeter and access-road lighting to a minimum uniform illuminance suitable for CCTV identification, dark-sky compliant fittings to limit spill beyond the boundary
CCTV	Perimeter camera coverage with no blind spot exceeding 5 m at the fence line, 90-day retention per PDC-SPEC-07 §3.4
Gatehouse	Security gatehouse and loading area, per PDC-BOD-001 §2.2 zone A–C/1–2

3.2 Drainage and stormwater

Item	Requirement
Foul water	Connection at the south-west corner chamber, per PDC-ER-001 §1.3
Surface water	Attenuated on-site before discharge, per Vantaa Water technical requirements and RIL drainage guidance
Stormwater discharge limit	Not exceeding 10 l/s/ha at the site boundary discharge point, for the 1-in-30-year storm event, with a 1-in-100-year event plus climate change allowance contained on-site without flooding a building
Attenuation method	Below-ground attenuation tanks or landscaped swales within the site's external areas, subject to S2 selection
Oil/silt interceptor	Fitted to all hardstanding and access road drainage upstream of attenuation, sized to the relevant hardstanding area
Generator compound bunding	Separate, sealed drainage for the bunded fuel storage area, with no cross-connection to the general site surface water system

3.3 Site access and haul road

Item	Requirement
Permanent access road	Single access from Tikkurilantie, shared with an operating logistics neighbour; deliveries restricted to 06:00–22:00 weekdays per PDC-ER-001 §11
Haul road	Single one-way haul road along gridline N, outboard of the cooling yard, per PDC-BOD-001 §2.3
Pavement	Designed for construction traffic loading during the works, reinstated to permanent finish at each Section's completion
Hoarding	Relocated progressively along the numbered gridline immediately east of the most recently completed hall

3.4 Fibre entry routes and meet-me room

Item	Requirement
Routes	Two physically diverse fibre entry routes into the site
Meet-me room	Termination panels for carrier fibre; carrier tails excluded from Contractor scope
Carrier	Infrastructure shall not favour or restrict any particular carrier

neutrality	
Route separation	The two routes shall not share a common duct, trench or building entry point for any part of their length within the site boundary; minimum horizontal separation of 15 m maintained between the two routes for their full length within the site, reducing to a minimum of 5 m only where site constraints are demonstrated to the Engineer and agreed in writing, and never sharing a single manhole, duct bank or building penetration
Building entry points	The two routes shall enter the building envelope at physically separate points not less than 20 m apart, each with independent fire-stopped penetration
Duct construction	Minimum two spare ducts per route reserved for future carrier use, marked and capped

4. Installation and workmanship

4.1 Perimeter fencing and intrusion detection shall be installed as a continuous, gap-free enclosure before any Section's hall substructure works commence in that Section's zone, to establish site security ahead of plant and material deliveries.

4.2 Stormwater attenuation infrastructure sized for the full site shall be installed progressively so that each Section's hardstanding and roof drainage is connected to attenuation capacity sufficient for that Section at handover, without relying on later Sections' infrastructure.

4.3 The two fibre entry routes shall be trenched, ducted and as-built surveyed independently, with the as-built survey demonstrating the minimum separation distances of Section 3.4 for the full length of each route within the site boundary.

4.4 External lighting shall be installed and commissioned to demonstrate no unacceptable light spill toward the north-east residential boundary, coordinated with the acoustic requirements of PDC-SPEC-06 §4.4.

4.5 Landscaping and boundary screening shall be installed to the approved planning permission conditions (PDC-PP-001), including any planting required to screen the chiller/dry-cooler yard from the north-east residential boundary.

4.6 All external works adjacent to an operating Section shall be carried out under a live-environment method statement per PDC-ER-001 §7, referencing the hoarding line in effect at the time of the works.

4.7 Any blasting required in the eastern third of the plot for external works foundations (fence posts, gatehouse, generator compound hardstanding) shall be carried out subject to the vibration limits and neighbour notification requirements of PDC-SS-001 and PDC-PP-001.

5. Control and sequences of operation

5.1 Perimeter and access control integration

External security systems (fence-line detection, ANPR gate, perimeter CCTV) shall integrate with the BMS/security network specified in PDC-SPEC-07, reporting alarms to the same NOC alarm hierarchy (PDC-SPEC-07 §5.2).

5.2 External lighting control

Mode	Trigger	Behaviour
Dusk-to-dawn baseline	Photocell/astronomical time clock	Perimeter and access-road lighting at baseline illuminance
Intrusion boost	Perimeter detection alarm	Affected zone lighting boosted to full illuminance, CCTV auto-recall
Maintenance override	Manual, logged	Full illuminance for maintenance access, time-limited

5.3 Drainage monitoring

Parameter	Setpoint / limit	Action on exceedance
Boundary discharge flow rate	≤ 10 l/s/ha (1-in-30-year event)	Alarm to FM team, review attenuation performance
Attenuation tank high level	Per manufacturer's design level	Alarm, inspect for blockage
Oil/silt interceptor high level	Per manufacturer's design level	Schedule interceptor service
Generator bund leak/level detection	Any detected liquid	Critical alarm, immediate investigation, no discharge to general drainage

5.4 Fibre route monitoring

Duct integrity and route separation shall be recorded in the as-built survey (Section 4.3) and referenced in the DCIM/BMS documentation set for the meet-me room, though the fibre routes themselves are passive infrastructure and do not report live status to the BMS.

6. Testing and commissioning

Level	Activity	External works scope
L1	Factory acceptance testing	Perimeter detection equipment, ANPR system, gate motors — witnessed functional test
L2	Site delivery and static inspection	Visual inspection of fencing, ducting, attenuation tanks before backfill
L3	Individual component start-up and functional test	Gate operation, perimeter detection zone-by-zone test, external lighting circuits
L4	Integrated systems testing, including black-building and failure-mode tests	Perimeter security integration test with PDC-SPEC-07 systems; drainage flow test against the boundary discharge limit
L5	Integrated load-bank testing to full Section load for 24 hours continuous	Not directly applicable to external works; external lighting and security systems monitored throughout the Section's L5 electrical test for stable operation

6.1 The fibre route as-built survey and separation-distance demonstration shall be completed and accepted before the Section's occupancy inspection (käyttöönottotarkastus).

6.2 Stormwater attenuation performance shall be demonstrated by calculation at S2, and by a witnessed flow test at the boundary discharge point at L4, confirming compliance with the 10 l/s/ha limit.

7. Submittals

7.1 The Subcontractor shall submit, at the design stages defined in PDC-ER-001 §10:

- site layout and haul road/hoarding sequencing plan, cross-referenced to PDC-BOD-001 §2.3;
- drainage strategy and stormwater attenuation calculation, demonstrating the 10 l/s/ha boundary discharge limit;
- perimeter security system schedule and CCTV coverage plan;
- fibre route design drawings, including the as-built separation-distance survey for each Section as it is completed;
- external lighting design, including light-spill modelling toward the north-east boundary;
- landscaping and boundary screening plan, cross-referenced to PDC-PP-001.

8. Safety, training and verification

8. Spares, training and warranties

8.1 The Subcontractor shall supply a spares schedule covering not less than two years of routine maintenance items (gate motor components, perimeter detection sensor heads, external luminaire gear) and one critical spare of each ANPR camera and gate control unit type.

8.2 The Subcontractor shall provide not less than 2 days of operator training per Section, covering perimeter security system operation, gate control and drainage system maintenance, with training records forming part of the S4 handover documentation.

8.3 Equipment warranties shall run for not less than 24 months from the relevant Section's Service Commencement date, or 30 months from delivery, whichever is later, and shall be assignable to the Employer's facilities management provider.

8.4 The Subcontractor shall maintain a landscaping establishment and maintenance regime for not less than 12 months following each Section's handover, including replacement of failed planting used for boundary screening.

End of PDC-SPEC-08 Rev A.

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